AN INVERSE LIMIT OF FINITE SOBER SPACES OVER A DIRECTED SET IS SPECTRAL

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The description of the limit of the underlying sets and of a basis of the topology by pullbacks of opens. Invoked inside the counted proof (source lines 4859-4862). Provides the basis of quasi-compact opens used to verify the spectral axioms.
- The statement that a closed irreducible subset of a finite sober space has a unique generic point. Invoked inside the counted proof (source lines 4869-4872). Produces the compatible family of generic points.

Source reference. Original lemma lemma-describe-limits: sources/stacks-topology/pinned-original-source.tex, source lines 2591–2591. Navigation pointer only: the counted text refers to the Stacks item lemma-describe-limits, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-limits: sources/stacks-topology/pinned-original-source.tex, source lines 2575–2575. Navigation pointer only: the counted text refers to the Stacks item lemma-limits, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *The inverse limit of a directed inverse system of finite sober topological spaces is a spectral topological space.*

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Proof. Let I be a directed set. Let X_i be an inverse system of finite sober spaces over I. Let $X = \lim X_i$ which exists by Lemma lemma-limits. As a set $X = \lim X_i$. Denote $p_i : X \to X_i$ the projection. Because I is directed we may apply Lemma lemma-describe-limits. A basis for the topology is given by the opens $p_i^{-1}(U_i)$ for $U_i \subset X_i$ open. Since an open covering of $p_i^{-1}(U_i)$ is in particular an open covering in the profinite topology, we conclude that $p_i^{-1}(U_i)$ is quasi-compact. Given $U_i \subset X_i$ and $U_j \subset X_j$, then $p_i^{-1}(U_i) \cap p_j^{-1}(U_j) = p_k^{-1}(U_k)$ for some $k \geq i, j$ and open $U_k \subset X_k$. Finally, if $Z \subset X$ is irreducible and closed, then $p_i(Z) \subset X_i$ is irreducible and therefore has a unique generic point ξ_i (because X_i is a finite sober topological space). Then $\xi = \lim \xi_i$ is a generic point of Z (it is a point of Z as Z is closed). This finishes the proof. $\square$